# Chun Hei Michael Chan

EE PhD Candidate @ EPFL — Neuro-X
Medical Image Processing Lab (MIP) - Swiss Federal Institute of Technology Lausanne (EPFL)
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## Field of Interests

My research focuses on the theory of **directed graph signal processing/neural networks**, **network-based statistical methods**, and applications in **network neuroscience**.

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## Education

**Swiss Federal Institute of Technology Lausanne**, Lausanne, Switzerland — 2023 - Present
PhD Candidate, Electrical Engineering
Thesis on Directed Graph Signal Processing and applications on Neuroimaging

**Swiss Federal Institute of Technology Lausanne**, Lausanne, Switzerland — 2020 - 2023
M.Sc., Data Science
Thesis (Medical Image Processing Lab at Swiss Federal Institute of Technology Lausanne)

**Ecole Polytechnique**, Palaiseau, France — 2017 - 2020
B.Sc., Mathematics and Computer Science
Thesis (Centre des Mathématiques Appliqueées at Ecole Polytechnique)

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## Research Experiences

**PhD Candidate - Medical Imaging Processing Lab** — 10/2023 - Present
Theoretical development of directed GSP and applications on brain connectomics
Advisor: Dimitri Van De Ville

**Project Student / Research Assistant - Medical Imaging Processing Lab** — 11/2022 - 09/2023
Brain organization revealed by functional connectivity gradients
Supervised by Elenor Morgenroth and Laura Vilaclara
Advisor: Dimitri Van De Ville

**Research Assistant - Persuasive Technology Lab** — 04/2022 - 02/2023
Design and prototyping for self-motivation questionnaire
Self determination theory (SDT) approach.
Advisor: Mauro Cherubini

**Project Student - Center for Biomedical Imaging** — 01/2022 - 06/2022
Diagnosis of hepatitis C from microstructure deformation and time exchange
GM and WM microstructure modelling by diffusion MRI through SANDI and WMTI
Time exchange estimation between GM neurites and extra-cellular space NEXI model
Supervised by Jessie Julie Mosso
Advisor: Cristina Cudalbu

**Project Student - Centre des Mathématiques Appliqueées École Polytechnique** 02/2020 - 05/2020

Operational Research: SDP solver

Approximation algorithm to maxcut problem - derandomization of rounding relaxation problem

Advisor: Giovanni Conforti

**Project Student - LEARN Center** 09/2019 - 12/2019

Flipped Learning Class: analysis of effect

Analysis of regularity and clickstream activity of students in an introductory Linear Algebra course at EPFL

Student features clustering while looking at clusters' representative value's correlation to grades

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## Industry Experiences

**Algorithm Team Intern - Intuitive Surgical Inc.** 07/2022 - 09/2022

POC Amplitude Pulsatile Component (APC) Maps in Endoscope stream of images

Extracting rPPG (POS method) and measuring proxy for perfusion index (lock in amplitude)

APC map and signal refinement through tracking and warping of patches

**CTO-AI Team Intern - Logitech International S.A** 09/2021 - 02/2022

Evaluation of improvement in robustness and accuracy of HR/HRV by improving hardware and modelling

Introducing FPS and Infra-red sensitive camera for motion and difficult light conditions

Attention models and 3d cnn in rPPG-robustness improvements and interpretability

**Computer Vision Research Intern - Besedo Content Moderation** 06/2020 - 09/2020

Watermarked image Filtering

Detect watermark with high-confidence to remove watermarked photos uploaded by users to retail websites

Transfer Learning of general object detection on top of other statistical model.

**Data Science Intern - SGI Venture Limited** 06/2019 - 08/2019

Pet Facial Recognition in low size population

General face detection algorithm (YOLO architecture) and LBPH feature comparison

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## Scientific Contributions

- **Chan, C. H. M.**, et al. "Graph Diffusion-Advection Operator for Directed Graph Signal Processing." arXiv preprint arXiv:2606.16306 (**2026**). (Submitted to **Transactions on Signal and Information Processing over Networks**)

- **Chan, C. H. M**, Cionca, A. , and Van De Ville, D. . "Statistical testing on directed graphs by surrogate data generation." arXiv preprint arXiv:2606.00758 (**2026**). (In Press for **Transactions on Signal and Information Processing over Networks**)

- **Chan, C. H. M**, et al. "Individual differences of cortical and subcortical emotion-informed functional gradients." bioRxiv (**2026**): 2026-02. (Submitted to **Human Brain Mapping**)

- **Chan, C. H. M**, Cionca, A., and Van De Ville, D. "Graph Signal Surrogate Generation for Statistical Testing of Covariance Structure on Directed Graphs" 2026 34th **European Signal Processing Conference** (EUSIPCO). IEEE, **2026**.

- Cionca, A., **Chan, C. H. M** et al. "Directed Brain Connectomics Revealed by Bicommunity Structure." bioRxiv (**2026**): 2026-02. (Submitted to **Science Advances**)

- Spencer, A.P.C., …, **Chan, C. H. M.** et al. "The microstructure-weighted human connectome: network properties and structure-function correlations across spatial scales." bioRxiv (**2026**): 2026-05. (Submitted to **Network Neuroscience**)

- Mosso, J., **Chan, C. H. M.**, et al. "In vivo diffusion MRS and MRI reveal cerebellar microstructure alterations in the rat developing brain during hepatic encephalopathy". (**2025**) (Submitted to **Imaging Neuroscience**)

- Asadi, S., …, **Chan, C. H. M.**, et al. "Non-invasive prediction of conduction velocities in the human brain from MRI-derived microstructure features at 7 Tesla." bioRxiv (**2025**): 2025-10. (Submitted to **Imaging Neuroscience**)

- **Chan, C. H. M.**, Cionca, A., and Van De Ville, D. . "Hilbert transform on graphs: Let there be phase." **IEEE Signal Processing Letters** (**2025**).
- Cionca, A., **Chan, C. H. M.**, and Van De Ville, D. . "Community detection for directed networks revisited using bimodularity." **Proceedings of the National Academy of Sciences** 122.35 (**2025**): e2500571122.
- David, F.*, **Chan, C. H. M.*** et al. "Deep Neural Encoder-Decoder Model to Relate fMRI Brain Activity with Naturalistic Stimuli." 2025 47th Annual **International Conference of the IEEE Engineering in Medicine and Biology Society** (EMBC). IEEE, **2025**.
- Cionca, A.*, **Chan, C. H. M.***, et al. "Community-Driven Signal Processing On Directed Brain Graphs." 2025 33rd **European Signal Processing Conference** (EUSIPCO). IEEE, **2025**.
- Arnéra, J., **Chan, C.H.M.**, and Mauro Cherubini. "Digital, analog, or hybrid: Comparing strategies to support self-reflection." Proceedings of the 2024 **ACM Designing Interactive Systems Conference**. **2024**.
- Dietler, Nicola, ..., **Chan, C.H.M.** et al. "A convolutional neural network segments yeast microscopy images with high accuracy." **Nature communications** 11.1 (**2020**): 5723.

Conference Abstracts: **GSP Workshop (2026) / OHBM (2023, 2025) / ABIM (2025) / MRS Workshop (2022)**

## Teaching

**EPFL Student Projects Supervision** 09/2023 - Present
Project Supervisor
12 Master level students supervised for their semester-long projects.

**Neural Signals and Signal Processing (NX-421)** 09/2023 - Present
Head Teaching Assistant
Creation and correction of projects and exercises for Master level students.
General handling of course logistics and exam preparation

**Image Processing I/II (MICRO-511/MICRO-512)** 09/2023 - Present
Teaching Assistant
Correction of projects and exercises for Master level students and exam testing.

## Award

- Conference Travel Grant, **ICASSP 2026**
- 1st Runner-up, Swiss Society for Bioengineering Student Award, **SSBE 2023**
- 2nd Runner-up, Data Analytics Competition, **Facebook Analytics 2020**
- Scholarship Holder, Undergraduate Studies, **Bourse d'excellence Campus France**

## Technical Skills

**Languages**: Python, C++, C, COQ
**Software & Tools**: Git, LaTeX, Jupyter/google collab, nvidia-gpu computing, docker, PySpark, Unix, Unity3D, Swift IOS dev
**Computational Methods**: transformers, adversarial attacks, object detections/segmentations, LSTM/RNN, and many others techniques applied to projects (most can be found on my personal github)